MPG No. 1.1.1 1 [Type I, triclinic] [G tensor]

* Rank 0 tensor.

$$[G]$$

$$G = \mathbb{G}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} G_x & G_y & G_z \end{bmatrix}$$

$$G_x = \mathbb{G}_x^{(1)}$$
$$G_y = \mathbb{G}_y^{(1)}$$
$$G_z = \mathbb{G}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} G_{xx} & G_{xy} & G_{xz} \\ G_{xy} & G_{yy} & G_{yz} \\ G_{xz} & G_{yz} & G_{zz} \end{bmatrix}$$

$$G_{xx} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)}$$
$$G_{xy} = \mathbb{G}_{xy}^{(1)}$$
$$G_{xz} = \mathbb{G}_{xz}^{(1)}$$
$$G_{yy} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(1)} - \mathbb{G}_v^{(1)}$$
$$G_{yz} = \mathbb{G}_{yz}^{(1)}$$
$$G_{zz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & G_{xy} & G_{xz} \\ -G_{xy} & 0 & G_{yz} \\ -G_{xz} & -G_{yz} & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{Q}_z^{(1)}$$
$$G_{xz} = -\mathbb{Q}_y^{(1)}$$
$$G_{yz} = \mathbb{Q}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} G_{xxx} & G_{xxy} & G_{xxz} \\ G_{yyx} & G_{yyy} & G_{yyz} \\ G_{zzx} & G_{zzy} & G_{zzz} \\ G_{yzx} & G_{yyz} & G_{zzy} \\ G_{xxz} & G_{yzx} & G_{zzx} \\ G_{xxy} & G_{yyx} & G_{yzx} \end{bmatrix}$$

$$G_{xxx} = 3\mathbb{G}_x^{(1)} + 2\mathbb{G}_x^{\alpha(1)}$$
$$G_{xxy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\beta(1)}$$
$$G_{xxz} = \mathbb{G}_z^{(1)} - \mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\beta(1)}$$
$$G_{yyx} = \mathbb{G}_x^{(1)} - \mathbb{G}_x^{\alpha(1)} + \mathbb{G}_x^{\beta(1)}$$
$$G_{yyy} = 3\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)}$$
$$G_{yyz} = \mathbb{G}_z^{(1)} - \mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\beta(1)}$$
$$G_{zzx} = \mathbb{G}_x^{(1)} - \mathbb{G}_x^{\alpha(1)} - \mathbb{G}_x^{\beta(1)}$$
$$G_{zzy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\beta(1)}$$
$$G_{zzz} = 3\mathbb{G}_z^{(1)} + 2\mathbb{G}_z^{\alpha(1)}$$
$$G_{yzx} = \mathbb{G}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxy} & G_{xxz} \\ G_{yyx} & 0 & G_{yyz} \\ G_{zzx} & G_{zzy} & 0 \\ G_{yzx} & -\frac{G_{yyz}}{2} & -\frac{G_{zzy}}{2} \\ -\frac{G_{xxz}}{2} & G_{zxy} & -\frac{G_{zzx}}{2} \\ -\frac{G_{xxy}}{2} & -\frac{G_{yyx}}{2} & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{xxy} = 2\mathbb{G}_y^{(2)} - 2\mathbb{Q}_{xz}^{(1)}$$
$$G_{xxz} = 2\mathbb{G}_z^{(2)} + 2\mathbb{Q}_{xy}^{(1)}$$
$$G_{yyx} = 2\mathbb{G}_x^{(2)} + 2\mathbb{Q}_{yz}^{(1)}$$
$$G_{yyz} = 2\mathbb{G}_z^{(2)} - 2\mathbb{Q}_{xy}^{(1)}$$
$$G_{zzx} = 2\mathbb{G}_x^{(2)} - 2\mathbb{Q}_{yz}^{(1)}$$
$$G_{zzy} = 2\mathbb{G}_y^{(2)} + 2\mathbb{Q}_{xz}^{(1)}$$
$$G_{yzx} = 3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$\frac{G_{yyz}}{2} = \mathbb{G}_z^{(2)} - \mathbb{Q}_{xy}^{(1)}$$
$$\frac{G_{zzy}}{2} = \mathbb{G}_y^{(2)} + \mathbb{Q}_{xz}^{(1)}$$
$$\frac{G_{xxz}}{2} = \mathbb{G}_z^{(2)} + \mathbb{Q}_{xy}^{(1)}$$
$$G_{zxy} = -3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$\frac{G_{zzx}}{2} = \mathbb{G}_x^{(2)} - \mathbb{Q}_{yz}^{(1)}$$
$$\frac{G_{xxy}}{2} = \mathbb{G}_y^{(2)} - \mathbb{Q}_{xz}^{(1)}$$
$$\frac{G_{yyx}}{2} = \mathbb{G}_x^{(2)} + \mathbb{Q}_{yz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & G_{yzy} & G_{yzz} \\ G_{zxx} & G_{zxy} & G_{zxz} \\ G_{xyx} & G_{xyy} & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$
$$G_{yzy} = \mathbb{G}_z^{(3)} + \mathbb{Q}_{xy}^{(2)}$$
$$G_{yzz} = -\mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$
$$G_{zxx} = -\mathbb{G}_z^{(3)} + \mathbb{Q}_{xy}^{(2)}$$
$$G_{zxy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$
$$G_{zxz} = \mathbb{G}_x^{(3)} + \mathbb{Q}_{yz}^{(2)}$$
$$G_{xyx} = \mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$
$$G_{xyy} = -\mathbb{G}_x^{(3)} + \mathbb{Q}_{yz}^{(2)}$$
$$G_{xyz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} G_{xxxx} & G_{xxyy} & G_{xxzz} & G_{xxyz} & G_{xxzx} & G_{xxxy} \\ G_{xxyy} & G_{yyyy} & G_{yyzz} & G_{yyyz} & G_{yyzx} & G_{yyxy} \\ G_{xxzz} & G_{yyzz} & G_{zzzz} & G_{zzyz} & G_{zzzx} & G_{zzxy} \\ G_{xxyz} & G_{yyyz} & G_{zzyz} & G_{yzyz} & G_{yzzx} & G_{yzxy} \\ G_{xxzx} & G_{yyzx} & G_{zzzx} & G_{yzzx} & G_{zxzx} & G_{zxxy} \\ G_{xxxy} & G_{yyxy} & G_{zzxy} & G_{yzxy} & G_{zxxy} & G_{xyxy} \end{bmatrix}$$

$$G_{xxxx} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)}$$
$$G_{xxyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)}$$
$$G_{xxzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)}$$
$$G_{xxyz} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)}$$
$$G_{xxzx} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)}$$
$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)}$$
$$G_{yyyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)}$$
$$G_{yyzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - \mathbb{G}_v^{(1)}$$
$$G_{yyyz} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)}$$
$$G_{yyzx} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)}$$
$$G_{yyxy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)}$$
$$G_{zzzz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)}$$
$$G_{zzyz} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)}$$
$$G_{zzzx} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)}$$
$$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)}$$
$$G_{yzyz} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - \mathbb{G}_v^{(2)}$$
$$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)}$$
$$G_{yzxy} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)}$$
$$G_{zxzx} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + \mathbb{G}_v^{(2)}$$
$$G_{zxxy} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(2)}$$
$$G_{xyxy} = \mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & G_{xxyy} & G_{xxzz} & G_{xxyz} & G_{xxzx} & G_{xxxy} \\ -G_{xxyy} & 0 & G_{yyzz} & G_{yyyz} & G_{yyzx} & G_{yyxy} \\ -G_{xxzz} & -G_{yyzz} & 0 & G_{zzyz} & G_{zzzx} & G_{zzxy} \\ -G_{xxyz} & -G_{yyyz} & -G_{zzyz} & 0 & G_{yzzx} & G_{yzxy} \\ -G_{xxzx} & -G_{yyzx} & -G_{zzzx} & -G_{yzzx} & 0 & G_{zxxy} \\ -G_{xxxy} & -G_{yyxy} & -G_{zzxy} & -G_{yzxy} & -G_{zxxy} & 0 \end{bmatrix}$$

$$G_{xxyy} = -2\mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$
$$G_{xxzz} = 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)}$$
$$G_{xxyz} = \mathbb{G}_{yz}^{(3)} - 4\mathbb{Q}_x^{\beta(1)}$$
$$G_{xxzx} = \mathbb{G}_{xz}^{(3)} - 2\mathbb{Q}_y^{(1)} + 2\mathbb{Q}_y^{\alpha(1)} + 2\mathbb{Q}_y^{\beta(1)}$$
$$G_{xxxy} = \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_z^{(1)} - 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)}$$
$$G_{yyzz} = 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$
$$G_{yyyz} = \mathbb{G}_{yz}^{(3)} + 2\mathbb{Q}_x^{(1)} - 2\mathbb{Q}_x^{\alpha(1)} + 2\mathbb{Q}_x^{\beta(1)}$$
$$G_{yyzx} = \mathbb{G}_{xz}^{(3)} - 4\mathbb{Q}_y^{\beta(1)}$$
$$G_{yyxy} = \mathbb{G}_{xy}^{(3)} - 2\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)}$$
$$G_{zzyz} = \mathbb{G}_{yz}^{(3)} - 2\mathbb{Q}_x^{(1)} + 2\mathbb{Q}_x^{\alpha(1)} + 2\mathbb{Q}_x^{\beta(1)}$$
$$G_{zzzx} = \mathbb{G}_{xz}^{(3)} + 2\mathbb{Q}_y^{(1)} - 2\mathbb{Q}_y^{\alpha(1)} + 2\mathbb{Q}_y^{\beta(1)}$$
$$G_{zzxy} = \mathbb{G}_{xy}^{(3)} - 4\mathbb{Q}_z^{\beta(1)}$$
$$G_{yzzx} = -\mathbb{Q}_z^{(1)} - 4\mathbb{Q}_z^{\alpha(1)}$$
$$G_{yzxy} = \mathbb{Q}_y^{(1)} + 4\mathbb{Q}_y^{\alpha(1)}$$
$$G_{zxxy} = -\mathbb{Q}_x^{(1)} - 4\mathbb{Q}_x^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} G_{yzyz} & G_{yzzx} & G_{yzxy} \\ G_{yzzx} & G_{zxzx} & G_{zxxy} \\ G_{yzxy} & G_{zxxy} & G_{xyxy} \end{bmatrix}$$

$$G_{yzyz} = \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(4)}$$
$$G_{yzzx} = 2\mathbb{G}_{xy}^{(4)}$$
$$G_{yzxy} = 2\mathbb{G}_{xz}^{(4)}$$
$$G_{zxzx} = \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} - 2\mathbb{G}_v^{(4)}$$
$$G_{zxxy} = 2\mathbb{G}_{yz}^{(4)}$$
$$G_{xyxy} = \mathbb{G}_0^{(3)} + 4\mathbb{G}_u^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & G_{yzzx} & G_{yzxy} \\ -G_{yzzx} & 0 & G_{zxxy} \\ -G_{yzxy} & -G_{zxxy} & 0 \end{bmatrix}$$

$$G_{yzzx} = -\mathbb{Q}_z^{(2)}$$
$$G_{yzxy} = \mathbb{Q}_y^{(2)}$$
$$G_{zxxy} = -\mathbb{Q}_x^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} G_{xxyz} & G_{xxzx} & G_{xxxy} \\ G_{yyyz} & G_{yyzx} & G_{yyxy} \\ G_{zzyz} & G_{zzzx} & G_{zzxy} \\ G_{yzyz} & G_{yzzx} & G_{yzxy} \\ G_{zxyz} & G_{zxzx} & G_{zxxy} \\ G_{xyyz} & G_{xyzx} & G_{xyxy} \end{bmatrix}$$

$$G_{xxyz} = \mathbb{Q}_x^{(3)} + 2\mathbb{Q}_x^{(4)} + 2\mathbb{Q}_x^{\alpha(2)}$$
$$G_{xxzx} = 2\mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(3)} - \mathbb{Q}_y^{\alpha(2)} - \mathbb{Q}_y^{\beta(2)}$$
$$G_{xxxy} = -2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(3)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)}$$
$$G_{yyyz} = -2\mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(3)} - \mathbb{Q}_x^{\alpha(2)} + \mathbb{Q}_x^{\beta(2)}$$
$$G_{yyzx} = \mathbb{Q}_y^{(3)} + 2\mathbb{Q}_y^{(4)} + 2\mathbb{Q}_y^{\alpha(2)}$$
$$G_{yyxy} = 2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(3)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)}$$
$$G_{zzyz} = 2\mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(3)} - \mathbb{Q}_x^{\alpha(2)} - \mathbb{Q}_x^{\beta(2)}$$
$$G_{zzzx} = -2\mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(3)} - \mathbb{Q}_y^{\alpha(2)} + \mathbb{Q}_y^{\beta(2)}$$
$$G_{zzxy} = \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{(4)} + 2\mathbb{Q}_z^{\alpha(2)}$$
$$G_{yzyz} = -3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$
$$G_{yzzx} = -\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(4)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)}$$
$$G_{yzxy} = \mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(4)} - \mathbb{Q}_y^{\alpha(2)} + \mathbb{Q}_y^{\beta(2)}$$
$$G_{zxyz} = \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(4)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)}$$
$$G_{zxzx} = 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$
$$G_{zxxy} = -\mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(4)} - \mathbb{Q}_x^{\alpha(2)} - \mathbb{Q}_x^{\beta(2)}$$
$$G_{xyyz} = -\mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(4)} - \mathbb{Q}_y^{\alpha(2)} - \mathbb{Q}_y^{\beta(2)}$$
$$G_{xyzx} = \mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(4)} - \mathbb{Q}_x^{\alpha(2)} + \mathbb{Q}_x^{\beta(2)}$$
$$G_{xyxy} = 2\mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} G_{yzxx} & G_{yzyy} & G_{yzzz} & G_{yzyz} & G_{yzzx} & G_{yzxy} \\ G_{zxxx} & G_{zxyy} & G_{zxzz} & G_{zxyz} & G_{zxzx} & G_{zxxy} \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & G_{xyyz} & G_{xyzx} & G_{xyxy} \end{bmatrix}$$

$$G_{yzxx} = \mathbb{Q}_x^{(5)} + 2\mathbb{Q}_x^{(6)} + 2\mathbb{Q}_x^{\alpha(3)}$$
$$G_{yzyy} = -2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)}$$
$$G_{yzzz} = 2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)}$$
$$G_{yzyz} = -3\mathbb{G}_u^{(6)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$
$$G_{yzzx} = \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)}$$
$$G_{yzxy} = -\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)}$$
$$G_{zxxx} = 2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)}$$
$$G_{zxyy} = \mathbb{Q}_y^{(5)} + 2\mathbb{Q}_y^{(6)} + 2\mathbb{Q}_y^{\alpha(3)}$$
$$G_{zxzz} = -2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)}$$
$$G_{zxyz} = -\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)}$$
$$G_{zxzx} = 3\mathbb{G}_u^{(6)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$
$$G_{zxxy} = \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)}$$
$$G_{xyxx} = -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)}$$
$$G_{xyyy} = 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)}$$
$$G_{xyzz} = \mathbb{Q}_z^{(5)} + 2\mathbb{Q}_z^{(6)} + 2\mathbb{Q}_z^{\alpha(3)}$$
$$G_{xyyz} = \mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)}$$
$$G_{xyzx} = -\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)}$$
$$G_{xyxy} = 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
G_{xxxx} & G_{xxyy} & G_{xxzz} & G_{xxyz} & G_{xxzx} & G_{xxxy} & G_{xxzy} & G_{xxxz} & G_{xxyx} \\
G_{yyxx} & G_{yyyy} & G_{yyzz} & G_{yyyz} & G_{yyzx} & G_{yyxy} & G_{yyzy} & G_{yyxz} & G_{yyyx} \\
G_{zzxx} & G_{zzyy} & G_{zzzz} & G_{zzyz} & G_{zzzx} & G_{zzxy} & G_{zzzy} & G_{zzxz} & G_{zzyx} \\
G_{yzxx} & G_{yzyy} & G_{yzzz} & G_{yzyz} & G_{yzzx} & G_{yzxy} & G_{yzzy} & G_{yzxz} & G_{yzyx} \\
G_{zxxx} & G_{zxyy} & G_{zxzz} & G_{zxyz} & G_{zxzx} & G_{zxxy} & G_{zxzy} & G_{zxxz} & G_{zxyx} \\
G_{xyxx} & G_{xyyy} & G_{xyzz} & G_{xyyz} & G_{xyzx} & G_{xyxy} & G_{xyzy} & G_{xyxz} & G_{xyyx}
\end{bmatrix}$$

$$G_{xxxx} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)}$$

$$G_{xxyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 2\mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{xxzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} - \mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{xxyz} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + \mathbb{G}_{yz}^{(3)} + \mathbb{Q}_x^{(3)} + 2\mathbb{Q}_x^{(4)} + 2\mathbb{Q}_x^{\alpha(2)} - 4\mathbb{Q}_x^{\beta(1)}$$

$$G_{xxzx} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} + 2\mathbb{G}_{xz}^{(5)} - 2\mathbb{Q}_y^{(1)} + \mathbb{Q}_y^{(3)} + 2\mathbb{Q}_y^{\alpha(1)} - \mathbb{Q}_y^{\alpha(2)} + 2\mathbb{Q}_y^{\beta(1)} - \mathbb{Q}_y^{\beta(2)}$$

$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} + 2\mathbb{Q}_z^{(1)} + \mathbb{Q}_z^{(3)} - 2\mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\alpha(2)} + 2\mathbb{Q}_z^{\beta(1)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{xxzy} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + \mathbb{G}_{yz}^{(3)} - \mathbb{Q}_x^{(3)} - 2\mathbb{Q}_x^{(4)} - 2\mathbb{Q}_x^{\alpha(2)} - 4\mathbb{Q}_x^{\beta(1)}$$

$$G_{xxxz} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)} - 2\mathbb{Q}_y^{(1)} - \mathbb{Q}_y^{(3)} + 2\mathbb{Q}_y^{\alpha(1)} + \mathbb{Q}_y^{\alpha(2)} + 2\mathbb{Q}_y^{\beta(1)} + \mathbb{Q}_y^{\beta(2)}$$

$$G_{xxyx} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} + 2\mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{(3)} - 2\mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\alpha(2)} + 2\mathbb{Q}_z^{\beta(1)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{yyxx} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} + 2\mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{yyyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)}$$

$$G_{yyzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{yyyz} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} + 2\mathbb{Q}_x^{(1)} + \mathbb{Q}_x^{(3)} - 2\mathbb{Q}_x^{\alpha(1)} - \mathbb{Q}_x^{\alpha(2)} + 2\mathbb{Q}_x^{\beta(1)} + \mathbb{Q}_x^{\beta(2)}$$

$$G_{yyzx} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + \mathbb{G}_{xz}^{(3)} + \mathbb{Q}_y^{(3)} + 2\mathbb{Q}_y^{(4)} + 2\mathbb{Q}_y^{\alpha(2)} - 4\mathbb{Q}_y^{\beta(1)}$$

$$G_{yyxy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - 2\mathbb{Q}_z^{(1)} + \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\alpha(2)} + 2\mathbb{Q}_z^{\beta(1)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{yyzy} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} + 2\mathbb{G}_{yz}^{(5)} + 2\mathbb{Q}_x^{(1)} - \mathbb{Q}_x^{(3)} - 2\mathbb{Q}_x^{\alpha(1)} + \mathbb{Q}_x^{\alpha(2)} + 2\mathbb{Q}_x^{\beta(1)} - \mathbb{Q}_x^{\beta(2)}$$

$$G_{yyxz} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + \mathbb{G}_{xz}^{(3)} - \mathbb{Q}_y^{(3)} - 2\mathbb{Q}_y^{(4)} - 2\mathbb{Q}_y^{\alpha(2)} - 4\mathbb{Q}_y^{\beta(1)}$$

$$G_{yyyx} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - 2\mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\alpha(2)} + 2\mathbb{Q}_z^{\beta(1)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{zzxx} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{zzyy} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} - \mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{zzzz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)}$$

$$G_{zzyz} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} + 2\mathbb{G}_{yz}^{(5)} - 2\mathbb{Q}_x^{(1)} + \mathbb{Q}_x^{(3)} + 2\mathbb{Q}_x^{\alpha(1)} - \mathbb{Q}_x^{\alpha(2)} + 2\mathbb{Q}_x^{\beta(1)} - \mathbb{Q}_x^{\beta(2)}$$

$$G_{zzzx} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)} + 2\mathbb{Q}_y^{(1)} + \mathbb{Q}_y^{(3)} - 2\mathbb{Q}_y^{\alpha(1)} - \mathbb{Q}_y^{\alpha(2)} + 2\mathbb{Q}_y^{\beta(1)} + \mathbb{Q}_y^{\beta(2)}$$

$$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{(4)} + 2\mathbb{Q}_z^{\alpha(2)} - 4\mathbb{Q}_z^{\beta(1)}$$

$$G_{zzzy} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} - 2\mathbb{Q}_x^{(1)} - \mathbb{Q}_x^{(3)} + 2\mathbb{Q}_x^{\alpha(1)} + \mathbb{Q}_x^{\alpha(2)} + 2\mathbb{Q}_x^{\beta(1)} + \mathbb{Q}_x^{\beta(2)}$$

$$G_{zzxz} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} + 2\mathbb{G}_{xz}^{(5)} + 2\mathbb{Q}_y^{(1)} - \mathbb{Q}_y^{(3)} - 2\mathbb{Q}_y^{\alpha(1)} + \mathbb{Q}_y^{\alpha(2)} + 2\mathbb{Q}_y^{\beta(1)} - \mathbb{Q}_y^{\beta(2)}$$

$$G_{zzyx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} - \mathbb{Q}_z^{(3)} - 2\mathbb{Q}_z^{(4)} - 2\mathbb{Q}_z^{\alpha(2)} - 4\mathbb{Q}_z^{\beta(1)}$$

$$G_{yzxx} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} - \mathbb{G}_{yz}^{(3)} + 4\mathbb{Q}_x^{\beta(1)}$$

$$G_{yzyy} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(3)} - 2\mathbb{Q}_x^{(1)} + 2\mathbb{Q}_x^{\alpha(1)} - 2\mathbb{Q}_x^{\beta(1)}$$

$$G_{yzzz} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(3)} + 2\mathbb{Q}_x^{(1)} - 2\mathbb{Q}_x^{\alpha(1)} - 2\mathbb{Q}_x^{\beta(1)}$$

$$G_{yzyz} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$

$$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} - \mathbb{Q}_z^{(1)} + \mathbb{Q}_z^{(4)} - 4\mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{yzxy} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(1)} + \mathbb{Q}_y^{(4)} + 4\mathbb{Q}_y^{\alpha(1)} - \mathbb{Q}_y^{\alpha(2)} + \mathbb{Q}_y^{\beta(2)}$$

$$G_{yzzy} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)} - \mathbb{Q}_{xyz}^{(2)}$$

$$G_{yzxz} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(5)} - \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{(4)} - 4\mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{yzyx} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(1)} - \mathbb{Q}_y^{(4)} + 4\mathbb{Q}_y^{\alpha(1)} + \mathbb{Q}_y^{\alpha(2)} - \mathbb{Q}_y^{\beta(2)}$$

$$G_{zxxx} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(3)} + 2\mathbb{Q}_y^{(1)} - 2\mathbb{Q}_y^{\alpha(1)} - 2\mathbb{Q}_y^{\beta(1)}$$

$$G_{zxyy} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} - \mathbb{G}_{xz}^{(3)} + 4\mathbb{Q}_y^{\beta(1)}$$

$$G_{zxzz} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(3)} - 2\mathbb{Q}_y^{(1)} + 2\mathbb{Q}_y^{\alpha(1)} - 2\mathbb{Q}_y^{\beta(1)}$$

$$G_{zxyz} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(1)} + \mathbb{Q}_z^{(4)} + 4\mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{zxzx} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$

$$G_{zxxy} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(5)} - \mathbb{Q}_x^{(1)} + \mathbb{Q}_x^{(4)} - 4\mathbb{Q}_x^{\alpha(1)} - \mathbb{Q}_x^{\alpha(2)} - \mathbb{Q}_x^{\beta(2)}$$

$$G_{zxzy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{(4)} + 4\mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{zxxz} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)} - \mathbb{Q}_{xyz}^{(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} G_{yzxx} & G_{yzyy} & G_{yzzz} & G_{yzyz} & G_{yzzx} & G_{yzxy} & G_{yzzy} & G_{yzxz} & G_{yzyx} \\ G_{zxxx} & G_{zxyy} & G_{zxzz} & G_{zxyz} & G_{zxzx} & G_{zxxy} & G_{zxzy} & G_{zxxz} & G_{zxyx} \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & G_{xyyz} & G_{xyzx} & G_{xyxy} & G_{xyzy} & G_{xyxz} & G_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
G_{yzxx} &= \mathbb{Q}_x^{(5)} + 2\mathbb{Q}_x^{(6)} + 2\mathbb{Q}_x^{\alpha(3)} \\
G_{yzyy} &= -2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{yzzz} &= 2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{yzyz} &= \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} - 3\mathbb{G}_u^{(6)} + 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{yzzx} &= 2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{yzxy} &= 2\mathbb{G}_{xz}^{(4)} - \mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{yzzy} &= -\mathbb{G}_0^{(3)} + 2\mathbb{G}_u^{(4)} - 3\mathbb{G}_u^{(6)} - 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{yzxz} &= -2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{yzyx} &= -2\mathbb{G}_{xz}^{(4)} - \mathbb{G}_{xz}^{(6)} - \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{zxxx} &= 2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{zxyy} &= \mathbb{Q}_y^{(5)} + 2\mathbb{Q}_y^{(6)} + 2\mathbb{Q}_y^{\alpha(3)} \\
G_{zxzz} &= -2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{zxyz} &= 2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxzx} &= \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} + 3\mathbb{G}_u^{(6)} - 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{zxxy} &= 2\mathbb{G}_{yz}^{(4)} + \mathbb{G}_{yz}^{(6)} - \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{zxzy} &= -2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxxz} &= -\mathbb{G}_0^{(3)} + 2\mathbb{G}_u^{(4)} + 3\mathbb{G}_u^{(6)} + 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{zxyx} &= -2\mathbb{G}_{yz}^{(4)} + \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{xyxx} &= -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{xyyy} &= 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{xyzz} &= \mathbb{Q}_z^{(5)} + 2\mathbb{Q}_z^{(6)} + 2\mathbb{Q}_z^{\alpha(3)} \\
G_{xyyz} &= 2\mathbb{G}_{xz}^{(4)} + \mathbb{G}_{xz}^{(6)} - \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{xyzx} &= 2\mathbb{G}_{yz}^{(4)} - \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{xyxy} &= \mathbb{G}_0^{(3)} + 4\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{xyzy} &= -2\mathbb{G}_{xz}^{(4)} + \mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{xyxz} &= -2\mathbb{G}_{yz}^{(4)} - \mathbb{G}_{yz}^{(6)} - \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{xyyx} &= -\mathbb{G}_0^{(3)} - 4\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} G_{xxxx} & G_{xxxy} & G_{xxxz} \\ G_{yyyx} & G_{yyyy} & G_{yyyz} \\ G_{zzzx} & G_{zzzy} & G_{zzzz} \\ G_{yyzx} & G_{yyzy} & G_{yyzz} \\ G_{zzxx} & G_{zzxy} & G_{zzxz} \\ G_{xxyx} & G_{xxyy} & G_{xxyz} \\ G_{yzzx} & G_{yzzy} & G_{yzzz} \\ G_{zxxx} & G_{zxxy} & G_{zxxz} \\ G_{xxyy} + G_{yyzz} - G_{yzzy} - G_{zxxz} + G_{zzxx} & G_{xyyy} & G_{xyyz} \\ -\frac{G_{yyyz}}{2} + \frac{G_{yyzy}}{2} + G_{zxxy} & G_{xyyz} + \frac{G_{zzzz}}{2} - \frac{G_{zzzz}}{2} & \frac{G_{xxxy}}{2} - \frac{G_{xxyx}}{2} + G_{xyyy} - G_{yyyx} + G_{zzxy} \end{bmatrix}$$

$$G_{xxxx} = 3\mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)}$$
$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)}$$
$$G_{xxxz} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)}$$
$$G_{yyyx} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)}$$
$$G_{yyyy} = 3\mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)}$$
$$G_{yyyz} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)}$$
$$G_{zzzx} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)}$$
$$G_{zzzy} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)}$$
$$G_{zzzz} = 3\mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)}$$
$$G_{yyzx} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + \mathbb{G}_{xz}^{(3)}$$
$$G_{yyzy} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} + 2\mathbb{G}_{yz}^{(5)}$$
$$G_{yyzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)}$$
$$G_{zzxx} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)}$$
$$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)}$$
$$G_{zzxz} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} + 2\mathbb{G}_{xz}^{(5)}$$
$$G_{xxyx} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)}$$
$$G_{xxyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 2\mathbb{G}_v^{(3)}$$
$$G_{xxyz} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + \mathbb{G}_{yz}^{(3)}$$
$$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)}$$
$$G_{yzzy} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)}$$
$$G_{yzzz} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(3)}$$
$$G_{zxxx} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(3)}$$
$$G_{zxxy} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(5)}$$
$$G_{zxxz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)}$$
$$G_{xyyy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)}$$
$$G_{xyyz} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(5)}$$